

Problem 8.1

Let

$$\hat{f}(n) := \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} f(x) dx$$

so that the Fourier series of a function f is

$$\sum_{n \in \mathbb{Z}} \hat{f}(n) e^{inx}.$$

Show that

$$\sum_{j=-N}^N e^{ijx} = \frac{\sin((N+1/2)x)}{\sin(x/2)}.$$

Proof. Let $f(x)$ be the function on the RHS. Notice that for any x on which $f(x)$ is defined, we have

$$f(x+2\pi) = \frac{\sin((N+1/2)x + (2N+1)\pi)}{\sin(x/2 + \pi)} = \frac{-\sin((N+1/2)x)}{-\sin(x/2)} = f(x)$$

that is, f is 2π -periodic where it is defined. In its current form, f is not defined at $2n\pi$ for $n \in \mathbb{Z}$, however, we have

$$\lim_{x \rightarrow 2n\pi} f(x) = \lim_{x \rightarrow 2n\pi} \frac{\sin((N+1/2)x)}{\sin(x/2)} = \lim_{x \rightarrow 2n\pi} \frac{(N+1/2) \cos((N+1/2)x)}{(1/2) \cos(x/2)} = 2N+1$$

so f can be extended to a 2π -periodic function on $\mathbb{R}$. This means that the Fourier series of f on $[-\pi, \pi]$ is valid on all of $\mathbb{R}$. We now compute that series. For any $n \in \mathbb{Z}$, we have

$$\begin{aligned} e^{-inx} f(x) &= e^{-inx} \frac{\sin((N+1/2)x)}{\sin(x/2)} \\ &= e^{-inx} \frac{e^{i(N+1/2)x} - e^{-i(N+1/2)x}}{e^{ix/2} - e^{-ix/2}} \\ &= e^{-inx} e^{-iNx} \frac{e^{i(2N+1)x} - 1}{e^{ix} - 1} \\ &= e^{-inx} e^{-iNx} \sum_{m=0}^{2N} e^{imx} \\ &= \sum_{m=-N}^N e^{i(m-n)x} \end{aligned}$$

For $m \neq n$, the term $e^{i(m-n)x}$ has an antiderivative $e^{i(m-n)x}/(i(m-n))$, which evaluates to 0 over the bounds $-\pi$ to π . Thus the n -th term of the Fourier series of f is

$$\hat{f}(n) = \begin{cases} \frac{1}{2\pi} \int_{-\pi}^{\pi} dx & \text{there is some } |m| \leq N, m = n \\ 0 & \text{otherwise} \end{cases} = \begin{cases} 1 & |n| \leq N \\ 0 & |n| \geq N \end{cases}$$

which corresponds to the Fourier series

$$f(x) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{inx} = \sum_{n=-N}^N e^{inx}$$

Therefore, proven. □

Problem 8.2

(a) Let $f(\theta) = |\theta|$ on $[-\pi, \pi]$.

(b) Show that

$$\hat{f}(n) = \begin{cases} \frac{\pi}{2}, & n = 0 \\ \frac{-1+(-1)^n}{\pi n^2}, & n \neq 0. \end{cases}$$

(c) Since $\sum_n |\hat{f}(n)| < \infty$, we will see in class that

$$f(\theta) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{in\theta}.$$

(d) Write the Fourier series of f in terms of sine and cosine.

(e) By taking $\theta = 0$, show that

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} &= \frac{\pi^2}{8}, \\ \sum_{n=1}^{\infty} \frac{1}{n^2} &= \frac{\pi^2}{6}. \end{aligned}$$

Note that

$$\sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \frac{1}{4} \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

Proof. (b). For $n = 0$, we have

$$\hat{f}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |x| e^{-i0x} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} |x| dx = \frac{\pi}{2}$$

and for $n \neq 0$, we have

$$\begin{aligned} \hat{f}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |x| e^{-inx} dx \\ &= \frac{1}{2\pi} \left(\int_0^{\pi} x e^{-inx} dx - \int_{-\pi}^0 x e^{-inx} dx \right) \\ &= \frac{1}{2\pi} \left(\int_0^{\pi} x e^{-inx} dx + \int_0^{\pi} x e^{inx} dx \right) \\ &= \frac{1}{\pi} \int_0^{\pi} x \cos(nx) dx \\ &= \frac{1}{\pi} \left(\frac{x \sin(nx)}{n} \Big|_0^{\pi} - \int_0^{\pi} \frac{\sin(nx)}{n} dx \right) \\ &= \frac{1}{\pi} \left(\frac{\cos(nx)}{n^2} \Big|_0^{\pi} \right) \\ &= \frac{-1 + (-1)^n}{\pi n^2} \end{aligned}$$

Thus the formula for $\hat{f}(n)$ is proven.

(d). By (c), we have

$$\begin{aligned}
f(\theta) &= \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{in\theta} \\
&= \sum_{n \in \mathbb{Z}} \hat{f}(n) \cos n\theta + i \sum_{n \in \mathbb{Z}} \hat{f}(n) \sin n\theta \\
&= \frac{\pi}{2} + \sum_{n \neq 0} \frac{-1 + (-1)^n}{\pi n^2} \cos n\theta + i \sum_{n \neq 0} \frac{-1 + (-1)^n}{\pi n^2} \sin n\theta \\
&= \frac{\pi}{2} + 2 \sum_{n=1}^{\infty} \frac{-1 + (-1)^n}{\pi n^2} \cos n\theta
\end{aligned}$$

(e). Substituting $\theta = 0$ to the above, we get

$$\sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^2} = \frac{\pi^2}{4}$$

The terms of the series above for even n are zero, so this just becomes

$$\sum_{n=1}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$$

We now decompose the absolutely convergent series $\sum_{n=1}^{\infty} 1/n^2$ in the following way

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{n=1}^{\infty} \frac{1}{(2n)^2} + \sum_{n=1}^{\infty} \frac{1}{(2n+1)^2} = \frac{1}{4} \sum_{n=1}^{\infty} \frac{1}{n^2} + \sum_{n=1}^{\infty} \frac{1}{(2n+1)^2}$$

We thus have the following:

$$\frac{3}{4} \sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{n=1}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8} \implies \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

□

Problem 8.3

Let $F_N(x)$ be the Fejér kernel from class,

$$F_N(x) = \frac{1}{N} \sum_{n=0}^{N-1} D_n(x), \quad D_N(x) = \sum_{n=-N}^N e^{inx}.$$

(a) Show that

$$F_N(x) = \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)}.$$

(b) Show that

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} F_N(x) dx = 1,$$

and that

$$\int_{|x| > \delta} F_N(x) dx \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

That is, $F_N(x)$ is an approximate identity in the sense discussed in class.

- (c) Suppose that $f : [-\pi, \pi] \rightarrow \mathbb{R}$ is continuous except at a point $\theta_0 \in (-\pi, \pi)$, where it has left and right limits $f(\theta_0^-)$ and $f(\theta_0^+)$ respectively. Show that

$$\lim_{N \rightarrow \infty} (f * F_N)(\theta_0) = \frac{f(\theta_0^+) + f(\theta_0^-)}{2},$$

where $(f * g)(x) := \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-y)g(y) dy$.

Proof. (a). Note that $D_n(x) = \sin((n+1/2)x)/\sin(x/2)$ by problem 1. We then have the following:

$$\begin{aligned} \sum_{n=0}^{N-1} D_n(x) &= \frac{1}{\sin(x/2)} \sum_{n=0}^{N-1} \sin((n+1/2)x) \\ &= \frac{1}{\sin(x/2)} \operatorname{Im} \left(\sum_{n=0}^{N-1} e^{i(n+1/2)x} \right) \\ &= \frac{1}{\sin(x/2)} \operatorname{Im} \left(e^{ix/2} \frac{e^{iNx} - 1}{e^{ix} - 1} \right) \\ &= \frac{1}{\sin(x/2)} \operatorname{Im} \left(e^{iNx/2} \frac{e^{iNx/2} - e^{-iNx/2}}{e^{ix/2} - e^{-ix/2}} \right) \\ &= \frac{1}{\sin(x/2)} \operatorname{Im} \left(e^{iNx/2} \frac{\sin(Nx/2)}{\sin(x/2)} \right) \\ &= \frac{\sin^2(Nx/2)}{\sin^2(x/2)} \end{aligned}$$

Hence the formula for F_N is proven.

(b). For the first one, we have

$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} F_N(x) dx &= \frac{1}{2N\pi} \sum_{n=0}^{N-1} \sum_{m=-n}^n \int_{-\pi}^{\pi} e^{imx} dx \\ &= \frac{1}{2N\pi} \sum_{n=0}^{N-1} \left(\int_{-\pi}^{\pi} dx + \sum_{m=-n, m \neq 0}^n \int_{-\pi}^{\pi} e^{imx} dx \right) \\ &= \frac{1}{2N\pi} \sum_{n=0}^{N-1} \left(2\pi + \sum_{m=-n, m \neq 0}^n \frac{e^{imx}}{im} \Big|_{-\pi}^{\pi} \right) \\ &= 1 \end{aligned}$$

Let $0 < \delta < \pi$. Notice that $F_N(x)$ is bounded above by $\frac{1}{N} \csc^2(x/2)$. $\csc$ is continuous on the compact subset $[-\pi, -\delta] \cup [\delta, \pi]$. The upper bound $\frac{1}{N} \csc^2(x/2)$ then converges to 0 uniformly so the integral (which is positive)

$$\int_{|x|>\delta} F_N(x) dx \rightarrow \int_{|x|>\delta} 0 dx = 0$$

(c). We first note that F_N is an even function, so $\frac{1}{2\pi} \int_0^{\pi} F_N(x) dx = \frac{1}{2}$. The convolution $(f * F_N)(\theta_0)$ is

$$(f * F_N)(\theta_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta_0 - y) F_N(y) dy$$

Given $\varepsilon > 0$, we take $\delta > 0$ such that $|f(\theta_0 - y) - f(\theta_0^+)| < \varepsilon$ for $y \in (-\delta, 0)$ and $|f(\theta_0 - y) - f(\theta_0^-)| < \varepsilon$ for $y \in (0, \delta)$. This prompts us to rewrite the convolution as

$$(f * F_N)(\theta_0) = \frac{1}{2\pi} \int_{|y| \geq \delta} f(\theta_0 - y) F_N(y) dy + \frac{1}{2\pi} \int_0^{\delta} f(\theta_0 - y) F_N(y) dy + \frac{1}{2\pi} \int_{-\delta}^0 f(\theta_0 - y) F_N(y) dy$$

Given a bound $M > 0$ for the function f , the first integral can be bounded between multiples of $\int_{|x|>\delta} F_N(x)dx$ which converges to 0 as $N \rightarrow \infty$. The second one can be bounded in the following way:

$$\begin{aligned} \frac{1}{2\pi} \int_0^\delta (f(\theta_0^-) - \varepsilon) F_N(y) dy &\leq \frac{1}{2\pi} \int_0^\delta f(\theta_0 - y) F_N(y) dy \leq \frac{1}{2\pi} \int_0^\delta (f(\theta_0^-) + \varepsilon) F_N(y) dy \\ \implies (f(\theta_0^-) - \varepsilon) \frac{1}{2\pi} \int_0^\delta F_N(y) dy &\leq \frac{1}{2\pi} \int_0^\delta f(\theta_0 - y) F_N(y) dy \leq (f(\theta_0^-) + \varepsilon) \frac{1}{2\pi} \int_0^\delta F_N(y) dy \end{aligned}$$

As $N \rightarrow \infty$, $\frac{1}{2\pi} \int_0^\delta F_N(y) dy \rightarrow \frac{1}{2}$ as a consequence of (b). We thus have

$$\frac{f(\theta_0^-) - \varepsilon}{2} \leq \frac{1}{2\pi} \int_0^\delta f(\theta_0 - y) F_N(y) dy \leq \frac{f(\theta_0^-) + \varepsilon}{2}$$

Similarly, we get the following bound for the third integral:

$$\frac{f(\theta_0^+) - \varepsilon}{2} \leq \frac{1}{2\pi} \int_0^\delta f(\theta_0 - y) F_N(y) dy \leq \frac{f(\theta_0^+) + \varepsilon}{2}$$

We thus have that $\lim_{N \rightarrow \infty} (f * F_N)(\theta_0)$ is bounded between $\frac{f(\theta_0^+) + f(\theta_0^-)}{2} - \varepsilon$ and $\frac{f(\theta_0^+) + f(\theta_0^-)}{2} + \varepsilon$, so taking $\varepsilon \rightarrow 0$, we get

$$\lim_{N \rightarrow \infty} (f * F_N)(\theta_0) = \frac{f(\theta_0^+) + f(\theta_0^-)}{2}$$

□

Problem 8.4

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be 2π -periodic such that $|f(x+h) - f(x)| \leq |h|^\alpha$ for all $x \in \mathbb{R}$ and $h \in \mathbb{R}$ for some $0 < \alpha < 1$. Show that

$$|\hat{f}(n)| \leq C|n|^{-\alpha}$$

for some $C > 0$.

Proof. For any $x \in \mathbb{R}$ and nonzero $k \in \mathbb{Z}$, we have

$$\begin{aligned} \pi^\alpha |k|^{-\alpha} &\geq |f(x + \pi/k) - f(x)| \\ &= \left| \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{in(x + \pi/k)} - \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{inx} \right| \\ &= \left| \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{inx} (e^{i\pi n/k} - 1) \right| \end{aligned}$$

Note the following integral identity:

$$\int_{-\pi}^{\pi} e^{-ikx} \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{inx} (e^{i\pi n/k} - 1) dx = \sum_{n \in \mathbb{Z}} \hat{f}(n) (e^{i\pi n/k} - 1) \int_{-\pi}^{\pi} e^{i(n-k)x} dx = -4\pi \hat{f}(k)$$

We can now bound $|\hat{f}(k)|$:

$$\begin{aligned}
|\hat{f}(k)| &= \frac{4}{\pi} \left| \int_{-\pi}^{\pi} e^{-ikx} \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{inx} (e^{i\pi n/k} - 1) dx \right| \\
&\leq \frac{4}{\pi} \int_{-\pi}^{\pi} \left| e^{-ikx} \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{inx} (e^{i\pi n/k} - 1) \right| dx \\
&\leq 4\pi^{\alpha-1} |k|^{-\alpha} \int_{-\pi}^{\pi} |e^{-ikx}| dx \\
&= 4\pi^{\alpha-1} |k|^{-\alpha} \int_{-\pi}^{\pi} dx \\
&= 8\pi^{\alpha} |k|^{-\alpha}
\end{aligned}$$

Therefore proven. □

Problem 8.5

Let

$$V_n(x) := (1 + e^{inx} + e^{-inx})F_n(x).$$

- (a) Check that $\hat{V}_n(k) = 1$ for $|k| \leq n$. Show that $\|V'_n\|_1 \leq Cn$ for some $C > 0$ independent of n .
- (b) Suppose that f is a trigonometric polynomial with $\hat{f}(k) = 0$ for $|k| > n$. Show that

$$\|f'\|_1 \leq Cn\|f\|_1.$$

Hint: Show that $f = V_n * f$, that $f' = V'_n * f$, and that $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$ for any f, g .

Proof. (a). For $|k| \leq n$, we have

$$\begin{aligned}
\hat{V}_n(k) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-ikx} (1 + e^{inx} + e^{-inx}) F_n(x) dx \\
&= \frac{1}{2n\pi} \sum_{m=0}^{n-1} \sum_{j=-m}^m \int_{-\pi}^{\pi} e^{-ikx} (1 + e^{inx} + e^{-inx}) e^{ijx} dx \\
&= \frac{1}{2n\pi} \sum_{m=0}^{n-1} \sum_{j=-m}^m \int_{-\pi}^{\pi} (e^{-i(k-j)x} + e^{i(n+j-k)x} + e^{-i(n-j+k)x}) dx \\
&= \frac{1}{n} \sum_{m=0}^{n-1} \sum_{j=-m}^m (\chi[j = k] + \chi[j = k - n] + \chi[j = k + n])
\end{aligned}$$

where $\chi[P]$, the indicator function, is 1 if P holds and 0 otherwise. We now try evaluating that sum for each

indicator separately:

$$\begin{aligned}
\sum_{m=0}^{n-1} \sum_{j=-m}^m \chi[j=k] &= \sum_{m=|k|}^{n-1} \sum_{j=-m}^m \chi[j=k] = \sum_{m=|k|}^{n-1} 1 = n - |k| \\
\sum_{m=0}^{n-1} \sum_{j=-m}^m \chi[j=k-n] &= \sum_{m=0}^{n-1} \sum_{j=n-m}^{n+m} \chi[j=k] \\
&= \begin{cases} \sum_{m=n-k}^{n-1} \sum_{j=n-m}^{n+m} \chi[j=k] & k > 0 \\ 0 & k \leq 0 \end{cases} \\
&= \begin{cases} k & k > 0 \\ 0 & k \leq 0 \end{cases} \\
\sum_{m=0}^{n-1} \sum_{j=-m}^m \chi[j=k+n] &= \sum_{m=0}^{n-1} \sum_{j=-n-m}^{-n+m} \chi[j=k] \\
&= \begin{cases} \sum_{m=n+k}^{n-1} \sum_{j=-n-m}^{-n+m} \chi[j=k] & k < 0 \\ 0 & k \geq 0 \end{cases} \\
&= \begin{cases} -k & k < 0 \\ 0 & k \geq 0 \end{cases}
\end{aligned}$$

Adding all of these, we obtain

$$\sum_{m=0}^{n-1} \sum_{j=-m}^m (\chi[j=k] + \chi[j=k-n] + \chi[j=k+n]) = n$$

and hence $\hat{V}_n(k) = 1$.

We write $V_n(x) = (1 + e^{inx} + e^{-inx}) \frac{\sin^2(nx/2)}{n \sin^2(x/2)}$ and compute its derivative:

$$V'_n(x) = (ine^{inx} - ine^{-inx})F_n(x) + (1 + e^{inx} + e^{-inx})F'_n(x)$$

Thus we have

$$\|V'_n\|_1 \leq 2n \|F_n\|_1 + 3 \|F'_n\|_1$$

Since $F_n(x) \geq 0$ (by Question 2) and $\int_{-\pi}^{\pi} F_n(x) dx = 2\pi$, we have $\|F_n\|_1 = 2\pi$. For the norm of F'_n , we note that F_n is a trigonometric polynomial with coefficients bounded by 1, we have that $|F'_n(x)| \leq n(n+1) \leq An^2$ for some $A > 0$, where A is independent of n . The derivative of F_n is

$$F'_n(x) = \frac{\sin(nx/2) \cos(nx/2)}{\sin^2(x/2)} - \frac{\sin^2(nx/2) \cos(x/2)}{n \sin^3(x/2)}$$

Since $|\sin x| \leq |x|$ for all x , $|\sin x| \geq \pi|x|/2$ for all $|x| < \pi/2$, and sine and cosine is bounded above by 1, we have

$$|F'_n(x)| \leq \frac{1}{\pi^2 x^2/4} + \frac{n|x|/2}{n\pi^3|x|^3/8} = \frac{4}{\pi^2 x^2} + \frac{4}{\pi^3 x^2} = \frac{B}{x^2}$$

for $x \in [-\pi, \pi]$ and some constant $B > 0$ independent of n . Then $\|F'_n\|_1$ can be bounded in the following

way

$$\begin{aligned}
\|F'_n\|_1 &= \int_{-\pi}^{\pi} |F'_n(x)| dx = \int_{|x| \leq 1/n} |F'_n(x)| dx + \int_{|x| \geq 1/n} |F'_n(x)| dx \\
&\leq \int_{|x| \leq 1/n} A n^2 dx + \int_{|x| \geq 1/n} \frac{B}{x^2} dx \\
&= 2An + \left. \frac{-2B}{x} \right|_{1/n}^{\pi} \\
&= 2(A+B)n + \frac{-2B}{\pi}
\end{aligned}$$

This means that for some constant $c > 0$ that depends only on A and B (i.e. c is independent of n), we have $\|F'_n\|_1 \leq cn$. We thus have the bound

$$\|V'_n\|_1 \leq 2n \|F_n\|_1 + 3 \|F'_n\|_1 \leq 4\pi n + 3cn = Cn$$

for a constant $C > 0$ independent of n .

(b). Write $f(x) = \sum_{k=-n}^n \hat{f}(k) e^{ikx}$. To show that $f = V_n * f$, it suffices to show that their Fourier coefficients are equal. For $|k| \leq n$, we have

$$\widehat{V_n * f}(k) = \hat{V}_n(k) \hat{f}(k) = \hat{f}(k)$$

and for $|k| > n$,

$$\widehat{V_n * f}(k) = \hat{V}_n(k) \hat{f}(k) = 0 = \hat{f}(k)$$

Thus $f = V_n * f$. Differentiating this equation, we get

$$\begin{aligned}
f'(x) &= \frac{d}{dx} \frac{1}{2\pi} \int_{-\pi}^{\pi} V_n(x-y) f(y) dy \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\partial}{\partial x} V_n(x-y) f(y) dy \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} V'_n(x-y) f(y) dy \\
&= V'_n * f(x)
\end{aligned}$$

thus $f' = V'_n * f$. The 1-norm of the right hand side of this can be bounded

$$\begin{aligned}
\|V'_n * f\|_1 &= \int_{-\pi}^{\pi} |V'_n * f(x)| dx \\
&= \int_{-\pi}^{\pi} \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} V'_n(x-y) f(y) dy \right| dx \\
&\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} |V'_n(x-y) f(y)| dy dx \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} |V'_n(x-y) f(y)| dx dy \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} \int_{-\pi-y}^{\pi-y} |V'_n(t) f(y)| dt dy \\
&\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} |V'_n(t) f(y)| dt dy \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} |V'_n(t)| dt \int_{-\pi}^{\pi} |f(y)| dy \\
&\leq \|V'_n\|_1 \|f\|_1
\end{aligned}$$

By part (a), we obtain the following bound for the norm of f' :

$$\|f'\|_1 \leq Cn \|f\|_1$$

□

Problem 8.6

Continuing with notation from the previous problem, also for any $s > 0$ define the “norm” (explain why this is not a norm):

$$\|f\|_{H^s}^2 := |\hat{f}(0)|^2 + \sum_{n \in \mathbb{Z}} |n|^{2s} |\hat{f}(n)|^2.$$

For $0 < \alpha < 1$ define

$$\|f\|_{C^\alpha} := \sup_{x,h} \frac{|f(x+h) - f(x)|}{|h|^\alpha} + \sup_x |f(x)|.$$

In this problem, we will prove a version of the Sobolev embedding theorem via a discrete version of Littlewood-Paley theory.

(a) Let

$$\varphi_j(x) = V_{2^{j-1}}(x) \left(e^{i3 \cdot 2^{j-1}x} + e^{-i3 \cdot 2^{j-1}x} \right).$$

Check that $\hat{\varphi}_j(n) = 1$ for $2^j \leq |n| \leq 2^{j+1}$ and that $\hat{\varphi}(0) = 0$, which is the same as

$$\int_{-\pi}^{\pi} \varphi_j(x) dx = 0.$$

(b) Show that

$$|\varphi_j(x)| \leq C \frac{1}{2^j} \min(2^{2j}, |x|^{-2}).$$

Hint: similar bounds hold for $F_n(x)$.

(c) Show that

$$\sum_{2^j \leq |n| \leq 2^{j+1}} |\hat{f}(n)|^2 \leq \int |(\varphi_j * f)(x)|^2 dx.$$

(d) Show that

$$|(\varphi_j * f)(x)| = \frac{1}{2\pi} \left| \int_{-\pi}^{\pi} \varphi_j(y) (f(x-y) - f(x)) dy \right| \leq C \|f\|_{C^\alpha} 2^{-\alpha j}$$

Conclude that

$$\sum_{2^j \leq |n| \leq 2^{j+1}} |\hat{f}(n)|^2 \leq C \|f\|_{C^\alpha}^2 2^{-2\alpha j}$$

(e) Let $0 < \alpha < 1$ and let $\beta < \alpha$. Show that there is a $C_{\alpha,\beta}$ s.t.

$$\|f\|_{H^\beta} \leq C_{\alpha,\beta} \|f\|_\alpha$$

This is a version of the *Sobolev embedding theorem*. The idea of splitting up the Fourier coefficients into dyadic blocks (of comparable frequencies) is known as Littlewood-Paley theory and has many applications.

(f) Conclude that if $f \in C^\alpha$ for $\alpha > \frac{1}{2}$, then

$$\sum_n |\hat{f}(n)| < \infty.$$

Proof. (a). Let $2^j \leq |n| \leq 2^{j+1}$. We now compute the Fourier coefficient $\hat{\varphi}_j(n)$:

$$\begin{aligned}\hat{\varphi}_j(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} V_{2^{j-1}}(x) \left(e^{i3 \cdot 2^{j-1}x} + e^{-i3 \cdot 2^{j-1}x} \right) dx \\ &= \hat{V}_{2^{j-1}}(n - 3 \cdot 2^{j-1}) + \hat{V}_{2^{j-1}}(n + 3 \cdot 2^{j-1})\end{aligned}$$

Note that $V_{2^{j-1}}$ is a trigonometric polynomial of degree $2^j - 1$. If $n > 0$ then $n + 3 \cdot 2^{j-1} \geq 2^j + 3 \cdot 2^{j-1} > 2^{j+1}$, so the coefficient $\hat{V}_{2^{j-1}}(n + 3 \cdot 2^{j-1}) = 0$. On the other hand, we have $2^j \leq n \leq 2^{j+1} \Rightarrow -2^{j-1} \leq n - 3 \cdot 2^{j-1} \leq 2^{j-1}$, so by Question 5, the coefficient $\hat{V}_{2^{j-1}}(n - 3 \cdot 2^{j-1}) = 1$. Hence $\hat{\varphi}_j(n) = 1$. The case $n < 0$ is handled almost identically.

For $n = 0$, we have

$$\hat{\varphi}_j(0) = \hat{V}_{2^{j-1}}(-3 \cdot 2^{j-1}) + \hat{V}_{2^{j-1}}(3 \cdot 2^{j-1})$$

and since $|3 \cdot 2^{j-1}| > 2^j$ and $V_{2^{j-1}}$ is a trigonometric polynomial of degree $2^{j-1} - 1$. Thus both of the coefficients in the sum above are 0. Hence, $\hat{\varphi}_j(0) = 0$.

(b). Notice that $|\varphi_j(x)|$ can be bounded above in the following way:

$$|\varphi_j(x)| \leq 6|F_{2^{j-1}}(x)|$$

$F_{2^{j-1}}$ is a trigonometric polynomial of degree $2^{j-1} - 1$ with coefficients bounded by 1, so $|F_{2^{j-1}}(x)| \leq 2^j$ (remember to take negative terms into account), hence

$$|\varphi_j(x)| \leq 6 \frac{1}{2^j} 2^{2j}$$

On the other hand, we have

$$|F_{2^{j-1}}(x)| = \frac{1}{2^{j-1}} \frac{\sin^2(2^{j-1}x/2)}{\sin^2(x/2)} \leq \frac{1}{2^{j-1}} \frac{1}{x^2/\pi^2} = \frac{\pi^2}{2^{j-1}} |x|^{-2}$$

Thus, we have

$$|\varphi_j(x)| \leq 12\pi^2 \frac{1}{2^j} |x|^{-2}$$

Hence, for some constant $C > 0$ (independent of j), we have $|\varphi_j(x)| \leq C \frac{1}{2^j} \min(2^{2j}, |x|^{-2})$.

(c). I'm speechless.

$$\begin{aligned}\int_{-\pi}^{\pi} |\varphi_j * f(x)|^2 dx &\geq \frac{1}{2\pi} \int_{-\pi}^{\pi} |\varphi_j * f(x)|^2 dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\sum_n \hat{\varphi}_j(n) \hat{f}_j(n) e^{inx} \right) \overline{\left(\sum_m \hat{\varphi}_j(m) \hat{f}_j(m) e^{imx} \right)} dx \\ &= \sum_{n,m} \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\hat{\varphi}_j(n) \hat{f}_j(n) \right) \overline{\left(\hat{\varphi}_j(m) \hat{f}_j(m) \right)} e^{i(n-m)x} dx \\ &= \sum_n \left(\hat{\varphi}_j(n) \hat{f}_j(n) \right) \overline{\left(\hat{\varphi}_j(n) \hat{f}_j(n) \right)} \\ &= \sum_n |\hat{\varphi}_j(n)|^2 |\hat{f}_j(n)|^2\end{aligned}$$

We then have

$$\sum_{2^j \leq |n| \leq 2^{j+1}} |\hat{f}(n)|^2 = \sum_{2^j \leq |n| \leq 2^{j+1}} |\hat{\varphi}_j(n)|^2 |\hat{f}(n)|^2 \leq \sum_n |\hat{\varphi}_j(n)|^2 |\hat{f}(n)|^2 \leq \int_{-\pi}^{\pi} |\varphi_j * f(x)|^2 dx$$

(d). Still speechless

$$\begin{aligned}
|(\varphi_j * f)(x)| &= \frac{1}{2\pi} \left| \int_{-\pi}^{\pi} \varphi_j(y) f(x-y) dy - f(x) \int_{-\pi}^{\pi} \varphi_j(y) dy \right| \\
&= \frac{1}{2\pi} \left| \int_{-\pi}^{\pi} \varphi_j(y) (f(x-y) - f(x)) dy \right| \\
&\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |\varphi_j(y)| \cdot |y|^\alpha \|f\|_\alpha dy \\
&= \frac{1}{\pi} \int_0^{1/2^j} |\varphi_j(y)| \cdot |y|^\alpha \|f\|_\alpha dy + \frac{1}{\pi} \int_{1/2^j}^{\pi} |\varphi_j(y)| \cdot |y|^\alpha \|f\|_\alpha dy \\
&\leq \frac{C \|f\|_\alpha}{2^j \pi} \int_0^{1/2^j} 2^{2j} |y|^\alpha dy + \frac{C \|f\|_\alpha}{2^j \pi} \int_{1/2^j}^{\pi} |y|^{\alpha-2} dy \\
&= \frac{C \|f\|_\alpha}{\pi} \frac{2^{-j\alpha}}{\alpha+1} + \frac{C \|f\|_\alpha}{2^j \pi} \left(\frac{\pi^{\alpha-1}}{\alpha-1} - \frac{2^{-j(\alpha-1)}}{\alpha-1} \right) \\
&= \frac{C \|f\|_\alpha}{\pi} \frac{2^{-j\alpha}}{\alpha+1} - \frac{C \|f\|_\alpha}{2^j \pi} \frac{\pi^{\alpha-1}}{1-\alpha} + \frac{C \|f\|_\alpha}{\pi} \frac{2^{-j\alpha}}{1-\alpha} \\
&\leq C' \|f\|_\alpha 2^{-j\alpha}
\end{aligned}$$

Thus

$$\begin{aligned}
\sum_{2^j \leq |n| \leq 2^{j+1}} |\hat{f}(n)|^2 &\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |\varphi_j * f(x)|^2 dx \\
&\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} C'^2 \|f\|_\alpha^2 2^{-2j\alpha} dx \\
&= C \|f\|_\alpha^2 2^{-2j\alpha}
\end{aligned}$$

(e). I no want write words.

$$\begin{aligned}
\|f\|_{H^\beta}^2 &= |\hat{f}(0)|^2 + \sum_{n \neq 0} |n|^{2\beta} |\hat{f}(n)|^2 \\
&\leq |\hat{f}(0)|^2 + \sum_{j=0}^{\infty} |2^{j+1}|^{2\beta} \sum_{2^j \leq |n| \leq 2^{j+1}} |\hat{f}(n)|^2 \\
&\leq |\hat{f}(0)|^2 + 2^{2\beta} \sum_{j=0}^{\infty} 2^{2j\beta} C \|f\|_\alpha^2 2^{-2j\alpha} \\
&= |\hat{f}(0)|^2 + 2^{2\beta} C \|f\|_\alpha^2 \sum_{j=0}^{\infty} (4^{\beta-\alpha})^j \\
&= |\hat{f}(0)|^2 + 2^{2\beta} C \|f\|_\alpha^2 \frac{1}{1-4^{\beta-\alpha}} \\
&\leq C_{\alpha,\beta}^2 \|f\|_\alpha^2
\end{aligned}$$

Thus

$$\|f\|_{H^\beta} \leq C_{\alpha,\beta} \|f\|_\alpha$$

(f). Let $\alpha > 1/2$ and pick $1/2 < \beta < \alpha$. Write $\sum_{n \neq 0} |\hat{f}(n)|$ as $\sum_{n \neq 0} |n|^{-\beta} \cdot |n|^\beta |\hat{f}(n)|$, we apply Holder's inequality to get

$$\left(\sum_{n \neq 0} |\hat{f}(n)| \right)^2 \leq \left(\sum_{n \neq 0} |n|^{-2\beta} \right) \left(\sum_{n \neq 0} |n|^{2\beta} |\hat{f}(n)|^2 \right) \leq \left(\sum_{n \neq 0} |n|^{-2\beta} \right) \|f\|_{H^\beta}^2 \leq \left(\sum_{n \neq 0} |n|^{-2\beta} \right) C_{\alpha,\beta}^2 \|f\|_\alpha^2$$

The series in the latter expression converges since $2\beta > 1$, thus $\sum_{n \neq 0} |\hat{f}(n)| < \infty$. □